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COMPUTER NETWORKS

Ex.No.12	Examine Network Address Translation (NAT) using Cisco Packet Tracer
Date:	

AIM

To configure and examine **Network Address Translation (NAT)** using **Cisco Packet Tracer** and verify communication between a private network and a public network.

Requirements

- Cisco Packet Tracer
- 1 Router (Cisco 1941)
- 2 Switches (Cisco 2960)
- 2 PCs
- Copper Straight-Through cables

Network Topology

Private Network Public Network

PC0 -------- Switch0 -------- Router -------- Switch1 -------- PC1
192.168.1.10 NAT Router 203.0.113.10
G0/0 Public PC
192.168.1.1
|
G0/1
203.0.113.1

IP Addressing

Device	Interface	IP Address	Subnet Mask
PC0	FastEthernet0	192.168.1.10	255.255.255.0
Router	G0/0	192.168.1.1	255.255.255.0
Router	G0/1	203.0.113.1	255.255.255.0
PC1	FastEthernet0	203.0.113.10	255.255.255.0

PROCEDURE

Step 1: Create the topology

1. Open **Cisco Packet Tracer**.
2. Place:
    - Two PCs.
    - Two 2960 switches.
    - One 1941 router.
3. Connect:
    - PC0 → Switch0.
    - Switch0 → Router G0/0.
    - Router G0/1 → Switch1.
    - Switch1 → PC1.
4. Use **Copper Straight-Through** cables.

Step 2: Configure PC0

Go to:

PC0 → Desktop → IP Configuration

Enter:

IP Address : 192.168.1.10
Subnet Mask : 255.255.255.0
Default Gateway : 192.168.1.1
Step 3: Configure PC1

Go to:

PC1 → Desktop → IP Configuration

Enter:

IP Address : 203.0.113.10
Subnet Mask : 255.255.255.0
Default Gateway : 203.0.113.1
Step 4: Configure the Router

Open:

Router → CLI

Configure the inside interface:

```
Router> enable
Router# configure terminal

Router(config)# interface gigabitEthernet 0/0
Router(config-if)# ip address 192.168.1.1 255.255.255.0
Router(config-if)# ip nat inside
Router(config-if)# no shutdown
Router(config-if)# exit
```

Configure the outside interface:

```
Router(config)# interface gigabitEthernet 0/1
Router(config-if)# ip address 203.0.113.1 255.255.255.0
Router(config-if)# ip nat outside
Router(config-if)# no shutdown
Router(config-if)# exit
```

Step 5: Configure NAT

Create an access list to identify the private network:

```
Router(config)# access-list 1 permit 192.168.1.0 0.0.0.255
```

Configure **PAT (NAT Overload)**:

```
Router(config)# ip nat inside source list 1 interface gigabitEthernet 0/1 overload
```

This allows multiple private IP addresses to share the router's outside IP address.

Step 6: Verify the Router Configuration

Execute:

```
Router# show ip interface brief
```

Expected output:

```
Interface              IP-Address      Status
GigabitEthernet0/0     192.168.1.1     up
GigabitEthernet0/1     203.0.113.1     up
```

Step 7: Test Connectivity

From **PC0**, open:

Desktop → Command Prompt

Enter:

```
ping 192.168.1.1
```

Expected:

Reply from 192.168.1.1: bytes=32 time<1ms TTL=255

Now test the public-side PC:

```
ping 203.0.113.10
```

Expected:

Reply from 203.0.113.10: bytes=32 time<1ms TTL=127

Step 8: Verify NAT Translation

Generate traffic from PC0 by pinging PC1:

PC0> ping 203.0.113.10

Then, on the router, enter:

Router# show ip nat translations

Sample output:

Pro Inside global Inside local Outside local Outside global
icmp 203.0.113.1:1 192.168.1.10:1 203.0.113.10:1 203.0.113.10:1

This shows that the **private IP address 192.168.1.10** has been translated to the router's **public IP address 203.0.113.1**.

Step 9: Verify NAT Statistics

Enter:

Router# show ip nat statistics

Sample output:

Total active translations: 1

Inside interfaces:
  GigabitEthernet0/0

Outside interfaces:
  GigabitEthernet0/1

Hits: 5
Misses: 0

The exact numbers may vary depending on the number of packets generated.

SAMPLE OUTPUT

PC0 pinging PC1:

PC> ping 203.0.113.10

Pinging 203.0.113.10 with 32 bytes of data:

Reply from 203.0.113.10: bytes=32 time<1ms TTL=127
Reply from 203.0.113.10: bytes=32 time<1ms TTL=127
Reply from 203.0.113.10: bytes=32 time<1ms TTL=127
Reply from 203.0.113.10: bytes=32 time<1ms TTL=127

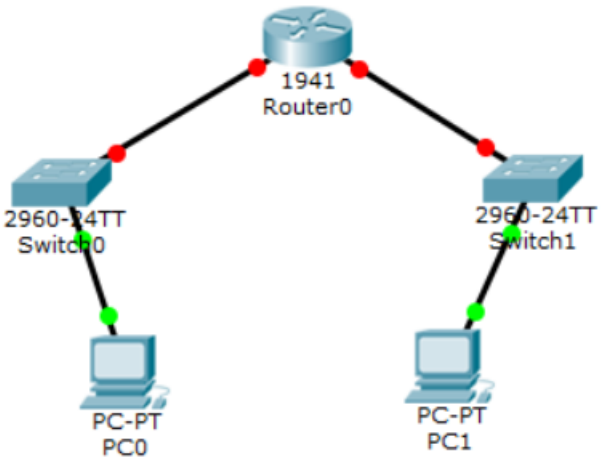
Ping statistics:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss)

NAT Translation Table:

Router# show ip nat translations

Pro	Inside global	Inside local	Outside local	Outside global
icmp	203.0.113.1:1	192.168.1.10:1	203.0.113.10:1	203.0.113.10:1

OUTPUT:



Router0

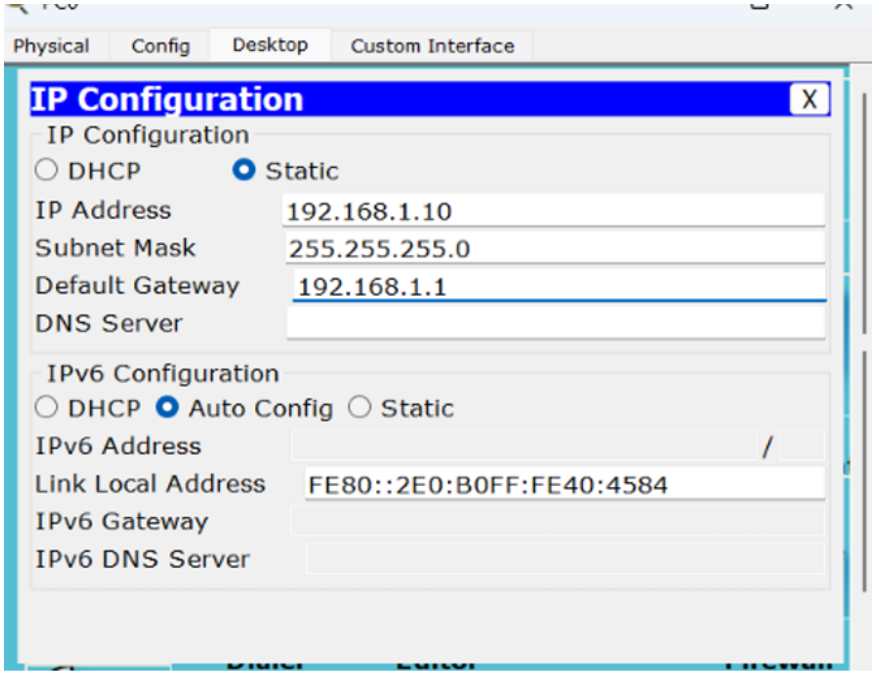
Physical | Config | CLI

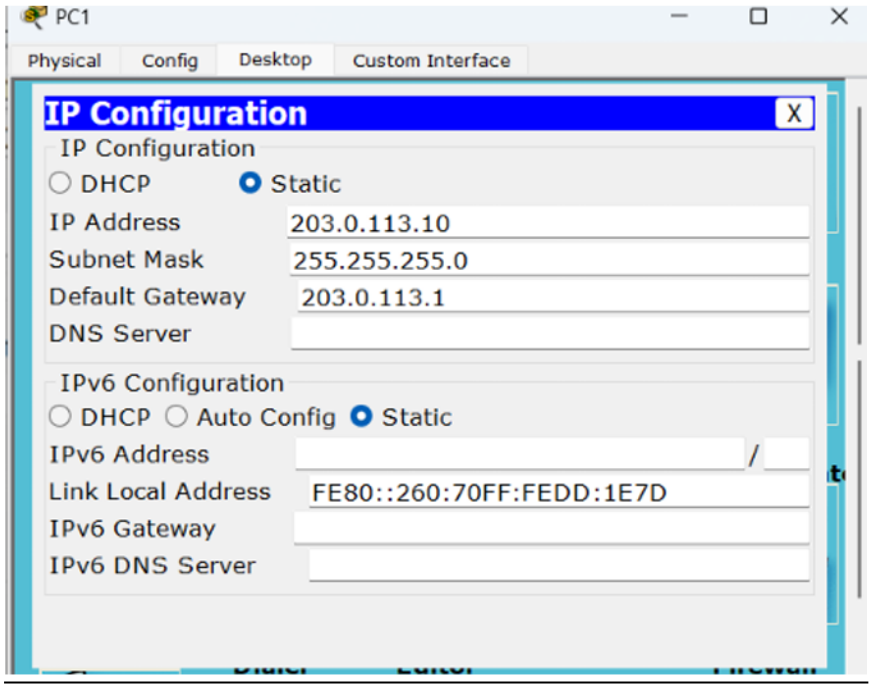
IOS Command Line Interface

```
ip nat inside source list 1 interface GigabitEthernet0/1 overload
ip classless
!
ip flow-export version 9
!
!
access-list 1 permit 192.168.1.0 0.0.0.255

Router#show ip nat translations
Pro  Inside global     Inside local       Outside local      Outside global
icmp 203.0.113.1:49    192.168.1.10:49    203.0.113.10:49    203.0.113.10:49
icmp 203.0.113.1:50    192.168.1.10:50    203.0.113.10:50    203.0.113.10:50
icmp 203.0.113.1:51    192.168.1.10:51    203.0.113.10:51    203.0.113.10:51
icmp 203.0.113.1:52    192.168.1.10:52    203.0.113.10:52    203.0.113.10:52
icmp 203.0.113.1:53    192.168.1.10:53    203.0.113.10:53    203.0.113.10:53
icmp 203.0.113.1:54    192.168.1.10:54    203.0.113.10:54    203.0.113.10:54
icmp 203.0.113.1:55    192.168.1.10:55    203.0.113.10:55    203.0.113.10:55
icmp 203.0.113.1:56    192.168.1.10:56    203.0.113.10:56    203.0.113.10:56

Router#show ip nat statistics
Total translations: 4 (0 static, 4 dynamic, 4 extended)
Outside Interfaces: GigabitEthernet0/1
Inside Interfaces: GigabitEthernet0/0
Hits: 8  Misses: 17
Expired translations: 4
Dynamic mappings:
```





RESULT

Thus, **Network Address Translation (NAT)** was successfully configured and examined using **Cisco Packet Tracer**. The private IP address **192.168.1.10** was translated into the public IP address **203.0.113.1**, and successful communication with the public network was verified.